Reading Notes: Analytic Combinatorics

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May 28, 2026

Contents

Chapter 1

COMBINATORIAL STRUCTURES AND ORDINARY GENERATING FUNCTIONS

1.1 Symbolic enumeration methods

$\triangleright$ **I.2.** *Unimodal permutations.* A unimodal permutation can be viewed as a sequence of elements increasing to a peak and then decreasing. This structure is represented by:

$$\mathcal{P}^{\mathcal{U}} = \textsc{Set}(\mathcal{Z}) \star \mathcal{Z}^{\blacksquare} \star \textsc{Set}(\mathcal{Z})$$

$$P^U(z) = \int_0^z (e^t \cdot \frac{d}{dt} t \cdot e^t) dt = \frac{1}{2}(e^{2z} - 1)$$

which yields

$$P_n^U = n![z^n]\frac{1}{2}e^{2z} = n!\frac{1}{2}\frac{2^n}{n!} = 2^{n-1}$$

Alternatively,

$$P_n^U = \sum_{k=1}^{n} \binom{n-1}{k-1} = \sum_{j=0}^{n-1} \binom{n-1}{j} = 2^{n-1}$$

where k is the position of n in the permutation.

1.2 Admissible constructions and specifications

1.3 Integer compositions and partitions

$\triangleright$ **I.16.** *Partitions with summands bounded in number and size.* The number of partitions of size n with at most k summands, each at most l, is

$$[z^n]\frac{(1-z)(1-z^2)\cdots(1-z^{k+l})}{((1-z)(1-z^2)\cdots(1-z^k))\cdot((1-z)(1-z^2)\cdots(1-z^l))} \tag{1.1}$$

Let $f(k,l)$ denote the generating function where the coefficient of z^n represents the number of such partitions. To derive a functional equation, consider the value of the largest part. There are two mutually exclusive possibilities: the largest part is strictly less than l, or it is exactly l. This distinction yields:

$$f(k,l) = f(k,l-1) + z^l f(k-1,l) \tag{1.2}$$

We can derive 1.1 from this relation by introducing $F_k(x)$, the generating function for $f(k,l)$ with respect to the index l:

$$F_k(x) = \sum_{l=0}^{\infty} f(k,l)x^l$$

For the base case $k = 0$, the only valid partition is the empty partition of size 0, implying $f(0,l) = 1$ for all $l \geq 0$. Thus:

$$F_0(x) = \sum_{l=0}^{\infty} 1 \cdot x^l = \frac{1}{1-x}$$

Summing 1.2 over $l \geq 1$ after multiplying by x^l, we obtain:

$$\sum_{l=1}^{\infty} f(k,l)x^l - \sum_{l=1}^{\infty} f(k,l-1)x^l = \sum_{l=1}^{\infty} f(k-1,l)(zx)^l$$

- The first term simplifies to $F_k(x) - 1$, since $f(k,0) = 1$.
- The second term is simply $xF_k(x)$.
- The right-hand side yields $F_{k-1}(zx) - 1$.

Plugging these in:

$$(1-x)F_k(x) - 1 = F_{k-1}(zx) - 1 \implies F_k(x) = \frac{1}{1-x}F_{k-1}(zx)$$

Now we apply this functional relation repeatedly:

$$\begin{aligned} F_k(x) &= \frac{1}{1-x}F_{k-1}(zx) \\ &= \frac{1}{1-x}\frac{1}{1-zx}F_{k-2}(z^2x) \\ &= \frac{1}{(1-x)(1-zx)\cdots(1-z^{k-1}x)}F_0(z^kx) \end{aligned}$$

Since $F_0(x) = \frac{1}{1-x}$, we have $F_0(z^kx) = \frac{1}{1-z^kx}$. Thus:

$$F_k(x) = \sum_{l=0}^{\infty} f(k,l)x^l = \frac{1}{(1-x)(1-zx)\cdots(1-z^kx)}$$

Since all the roots of the denominator are distinct (assuming $|z| < 1$ or that z is a formal variable), we can decompose this into a sum of partial fractions:

$$\frac{1}{\prod_{i=0}^{k}(1-z^ix)} = \sum_{j=0}^{k} \frac{A_j}{1-z^jx}$$

To find a specific A_j, we multiply both sides by $(1 - z^j x)$ and then set $x = z^{-j}$:

$$\begin{aligned} A_j &= \left[\frac{1 - z^j x}{\prod_{i=0}^{k} (1 - z^i x)} \right]_{x=z^{-j}} \\ &= \frac{1}{\prod_{i=0}^{j-1} (1 - z^i z^{-j}) \prod_{i=j+1}^{k} (1 - z^i z^{-j})} \\ &= \frac{(-1)^j z^{j(j+1)/2}}{\prod_{i=1}^{j} (1 - z^i) \prod_{i=1}^{k-j} (1 - z^i)} \end{aligned}$$

The coefficient of x^l in $F_k(x)$ is the sum of the coefficients of x^l in each partial fraction. For a term $\frac{A_j}{1-z^j x}$, the coefficient is $A_j (z^j)^l$.

$$\begin{aligned} f(k,l) &= \sum_{j=0}^{k} A_j (z^j)^l \\ &= \sum_{j=0}^{k} \frac{(-1)^j z^{j(j+1)/2} \cdot z^{jl}}{\prod_{i=1}^{j} (1 - z^i) \prod_{i=1}^{k-j} (1 - z^i)} \\ &= \frac{1}{\prod_{i=1}^{k} (1 - z^i)} \sum_{j=0}^{k} \binom{k}{j}_z (-1)^j z^{j(j+1)/2 + jl} \end{aligned}$$

where $\binom{k}{j}_z$ is the Gaussian binomial coefficient:

$$\begin{aligned} \binom{k}{j}_z &= \frac{\prod_{i=1}^{k} (1 - z^i)}{\prod_{i=1}^{j} (1 - z^i) \prod_{i=1}^{k-j} (1 - z^i)} \\ &= \frac{(1 - z^k)(1 - z^{k-1}) \cdots (1 - z^{k-j+1})}{(1 - z)(1 - z^2) \cdots (1 - z^j)} \end{aligned}$$

We can rewrite the exponent in the sum:

$$\frac{j(j+1)}{2} + jl = \frac{j(j-1)}{2} + j(l+1)$$

Now the sum becomes

$$f(k,l) = \frac{1}{\prod_{i=1}^{k} (1 - z^i)} \sum_{j=0}^{k} \binom{k}{j}_z (-1)^j z^{j(j-1)/2} (z^{l+1})^j$$

The q-binomial theorem states that for a variable X:

$$\prod_{i=0}^{k-1} (1 - z^i X) = \sum_{j=0}^{k} \binom{k}{j}_z (-1)^j z^{j(j-1)/2} X^j \tag{1.3}$$

Plugging $X = z^{l+1}$ into the theorem yields 1.1:

$$\begin{aligned} f(k,l) &= \frac{(1 - z^{l+1})(1 - z^{l+2}) \cdots (1 - z^{l+k})}{(1 - z^1)(1 - z^2) \cdots (1 - z^k)} \\ &= \frac{(1 - z)(1 - z^2) \cdots (1 - z^{k+l})}{((1 - z)(1 - z^2) \cdots (1 - z^k)) \cdot ((1 - z)(1 - z^2) \cdots (1 - z^l))} \end{aligned}$$

To prove 1.3, let $P_k(X)$ be the product we are expanding:

$$P_k(X) = \prod_{i=0}^{k-1}(1 - z^i X)$$

We want to show that $P_k(X) = \sum_{j=0}^{k} \binom{k}{j}_z (-1)^j z^{j(j-1)/2} X^j$. Consider the relationship between $P_k(X)$ and $P_{k-1}(X)$. By definition:

$$P_k(X) = (1 - z^{k-1}X)P_{k-1}(X)$$

Now notice what happens if we replace X with zX in $P_k(X)$:

$$P_k(zX) = \prod_{i=0}^{k-1}(1 - z^{i+1}X) = \frac{1 - z^k X}{1 - X} P_k(X)$$

Rearranging this gives us a functional equation for $P_k(X)$:

$$(1 - X)P_k(zX) = (1 - z^k X)P_k(X)$$

Let $P_k(X) = \sum_{j=0}^{k} c_j X^j$. We substitute this power series into the functional equation:

$$(1 - X)\sum_{j=0}^{k} c_j (zX)^j = (1 - z^k X)\sum_{j=0}^{k} c_j X^j$$

Now, we compare the coefficients of X^j on both sides. For a specific power j:

$$c_j z^j - c_{j-1} z^{j-1} = c_j - c_{j-1} z^k$$

Rearranging to solve for c_j in terms of c_{j-1}:

$$c_j = c_{j-1}\frac{z^{j-1} - z^k}{z^j - 1} = -c_{j-1} z^{j-1} \frac{1 - z^{k-j+1}}{1 - z^j}$$

Starting from the base case $c_0 = P_k(0) = 1$, we can find c_j by iteration:

$$c_1 = -c_0 z^0 \frac{1 - z^k}{1 - z^1}, \quad c_2 = -c_1 z^1 \frac{1 - z^{k-1}}{1 - z^2} = (-1)^2 z^{0+1} \frac{(1 - z^k)(1 - z^{k-1})}{(1 - z^1)(1 - z^2)}$$

The exponent of z becomes $0 + 1 + 2 + \cdots + (j - 1) = \frac{j(j-1)}{2}$. The fraction is the definition of the Gaussian binomial coefficient $\binom{k}{j}_z$. Thus, the coefficient is:

$$c_j = (-1)^j z^{j(j-1)/2} \binom{k}{j}_z$$

Plugging the coefficients back into the sum, we obtain:

$$\prod_{i=0}^{k-1}(1 - z^i X) = \sum_{j=0}^{k} \binom{k}{j}_z (-1)^j z^{j(j-1)/2} X^j$$

This completes the proof.

▷ **I.17.** *Systems of linear Diophantine inequalities.*

$$\{x_1 \geq 0,\ x_2 \geq 2x_1,\ x_3 \geq 2x_2,\ x_4 \geq 2x_3\}$$

We introduce non-negative slack variables $y_j \in \mathbb{Z}_{\geq 0}$ to quantify the "excess" of each term relative to its lower bound.

$$x_1 = y_1 \tag{1.4}$$
$$x_2 = 2x_1 + y_2 = 2y_1 + y_2 \tag{1.5}$$
$$x_3 = 2x_2 + y_3 = 4y_1 + 2y_2 + y_3 \tag{1.6}$$
$$x_4 = 2x_3 + y_4 = 8y_1 + 4y_2 + 2y_3 + y_4 \tag{1.7}$$

The total weight is $S = x_1 + x_2 + x_3 + x_4 = 15y_1 + 7y_2 + 3y_3 + y_4$. Since y_j are independent and range freely over non-negative integers, the OGF $F(z)$ is the product of the individual generating functions for each y_j:

- For $15y_1 : \sum_{y_1=0}^{\infty}(z^{15})^{y_1} = \frac{1}{1-z^{15}}$
- For $7y_2 : \sum_{y_2=0}^{\infty}(z^{7})^{y_2} = \frac{1}{1-z^{7}}$
- For $3y_3 : \sum_{y_3=0}^{\infty}(z^{3})^{y_3} = \frac{1}{1-z^{3}}$
- For $y_4 : \sum_{y_4=0}^{\infty}(z^{1})^{y_4} = \frac{1}{1-z^{1}}$

Multiplying these together yields the desired OGF:

$$F(z) = \frac{1}{(1-z)(1-z^3)(1-z^7)(1-z^{15})}$$

$$\{x_1 + x_2 \leq x_3\}$$

Following the same approach as in the previous problem, we use slack variables to convert the inequality into an equality of independent parts. The constraint $x_3 \geq x_1 + x_2$ can be rewritten by introducing a non-negative integer $y \geq 0$:

$$x_3 = x_1 + x_2 + y$$

The size of the composition is $S = x_1 + x_2 + x_3 + x_4 = 2x_1 + 2x_2 + y + x_4$. Since the variables x_1, x_2, y, x_4 are independent and each ranges freely over the non-negative integers, the OGF is the product of the generating functions for the individual terms in the sum:

- For $2x_1 : \frac{1}{1-z^2}$
- For $2x_2 : \frac{1}{1-z^2}$
- For $y : \frac{1}{1-z}$
- For $x_4 : \frac{1}{1-z}$

Multiplying these together gives:

$$F(z) = \frac{1}{(1-z^2)^2(1-z)^2}$$

$$\{x_1 + x_2 \geq x_3\}$$

Since the inequality does not immediately suggest a single slack variable, a more robust approach uses the complement method. Let $U(z) = (1-z)^{-4}$ be the OGF for four unrestricted variables. The universe of compositions partitions into those where $x_1+x_2 < x_3$ and our target class $x_1 + x_2 \geq x_3$. We find the OGF for the complement $x_1 + x_2 < x_3$. Set $x_3 = x_1 + x_2 + 1 + y$, where $y \geq 0$. The total sum S is:

$$S = x_1 + x_2 + (x_1 + x_2 + 1 + y) + x_4 = 2x_1 + 2x_2 + y + x_4 + 1$$

The OGF for this complement, which we denote by $C(z)$, is:

$$C(z) = \frac{z}{(1-z^2)^2(1-z)^2}$$

The OGF for our constraint is then:

$$F(z) = U(z) - C(z) = \frac{1}{(1-z)^4} - \frac{z}{(1-z^2)^2(1-z)^2} = \frac{1+z+z^2}{(1-z)^4(1+z)^2}$$

$$\{x_1 + x_2 \leq x_3 + x_4\}$$

To find the OGF for $\{x_1 + x_2 \leq x_3 + x_4\}$, we can use the complement method combined with the symmetry of the variables. Let $A = x_1 + x_2$ and $B = x_3 + x_4$. We seek the OGF for compositions satisfying $A \leq B$. In the universe of all possible compositions (x_1, x_2, x_3, x_4), there are three mutually exclusive cases: 1. $A < B$, 2. $A > B$, 3. $A = B$. By symmetry, the OGF for case 1 must be identical to the OGF for case 2, since A and B are built from the same number of variables with the same weights. Denote this common OGF by $G_<(z)$, and let $G_=(z)$ be the OGF for case 3. The total unrestricted OGF is $U(z) = \frac{1}{(1-z)^4}$. Therefore:

$$U(z) = 2G_<(z) + G_=(z)$$

Our target OGF is $F(z) = G_<(z) + G_=(z)$. Solving the system gives:

$$F(z) = \frac{U(z) + G_=(z)}{2}$$

Now $G_=(z)$ is the OGF for compositions in which the sum of the first two variables equals the sum of the last two.

- The number of compositions with $x_1 + x_2 = n$ is $[z^n]\frac{1}{(1-z)^2} = n + 1$.
- The number of compositions with $x_3 + x_4 = n$ is the same.
- For the equality $x_1 + x_2 = x_3 + x_4$ to hold, both sums must equal some $n \geq 0$.

$$G_=(z) = \sum_{n=0}^{\infty}(n+1)^2 z^{2n} = \frac{1+z^2}{(1-z^2)^3}$$

Substituting $U(z)$ and $G_=(z)$ into our formula yields:

$$F(z) = \frac{1}{2}\left[\frac{1}{(1-z)^4} + \frac{1+z^2}{(1-z^2)^3}\right] = \frac{1+z+2z^2}{(1-z)^4(1+z)^3}$$

$$\{x_1 \leq x_2,\ x_2 \geq x_3,\ x_3 \leq x_4\}$$

Because the inequalities alternate direction (up, down, up), we cannot use a straightforward chain of slack variable substitutions as we did for $x_1 \leq x_2 \leq x_3 \leq x_4$. Instead, we apply the inclusion–exclusion principle to the constraints. We begin with the universe of all unrestricted compositions, whose OGF is $U(z) = \frac{1}{(1-z)^4}$. We wish to enforce three conditions:

- $C_1 : x_1 \leq x_2$
- $C_2 : x_2 \geq x_3$
- $C_3 : x_3 \leq x_4$

However, it is simpler to work with chains of "less than or equal" inequalities. Consider the complement of the downward middle step. Take the set of all compositions satisfying both $x_1 \leq x_2$ and $x_3 \leq x_4$. From this set, we subtract those compositions that violate the middle constraint (i.e., $x_2 < x_3$).
The OGF for the set defined by $x_1 \leq x_2$ and $x_3 \leq x_4$: These two pairs are independent.

- For $x_1 \leq x_2$, set $x_2 = x_1 + y_1$. The total contribution is $2x_1 + y_1$, so the OGF is $\frac{1}{(1-z^2)(1-z)}$.
- For $x_3 \leq x_4$, the OGF is identically $\frac{1}{(1-z^2)(1-z)}$.

Since the two pairs are independent, we multiply their OGFs:

$$A(z) = \frac{1}{(1-z^2)^2(1-z)^2}$$

Now we find the OGF for the violation $x_1 \leq x_2 < x_3 \leq x_4$ by introducing slack variables:

- $x_1 = y_1$
- $x_2 = y_1 + y_2$
- $x_3 = y_1 + y_2 + 1 + y_3$
- $x_4 = y_1 + y_2 + 1 + y_3 + y_4$

Summing the four expressions gives:

$$S = 4y_1 + 3y_2 + 2y_3 + y_4 + 2$$

The OGF for this violation is therefore:

$$V(z) = \frac{z^2}{(1-z^4)(1-z^3)(1-z^2)(1-z)}$$

The OGF for the zigzag constraint is the OGF for the two independent pairs minus the OGF for the cases where the peak at x_2 fails:

$$\begin{aligned}F(z) &= \frac{1}{(1-z^2)^2(1-z)^2} - \frac{z^2}{(1-z^4)(1-z^3)(1-z^2)(1-z)} \\ &= \frac{1}{(1-z)(1-z^2)}\left[\frac{1}{(1-z)(1-z^2)} - \frac{z^2}{(1-z^3)(1-z^4)}\right]\end{aligned}$$

▷ **I.18.** *Odd versus distinct summands.* We establish a one-to-one correspondence between partitions into odd parts and partitions into distinct parts.

- Odd → Distinct
  1. Group identical odd parts together.
  2. If an odd part k appears m times, write m in binary: $m = 2^{a_1} + 2^{a_2} + \cdots$ with distinct powers of two.
  3. Replace the m copies of k by $2^{a_1} \cdot k,\ 2^{a_2} \cdot k,\ \ldots$
- Distinct → Odd
  1. Factor each part as $2^a \times$ odd.
  2. Split it into 2^a copies of that odd factor.
  3. Do this for all parts; now all parts are odd.
  4. Sort the resulting parts to obtain a partition into odd parts.

As an example:

$$\begin{aligned}3+3+3+3+3+1+1 &= 5 \times 3 + 2 \times 1 \\ &= (2^2 + 2^0) \times 3 + 2^1 \times 1 \\ &= 4 \times 3 + 1 \times 3 + 2 \times 1 \\ &= 12 + 3 + 2\end{aligned}$$

$$\begin{aligned}12+3+2 &= 4 \times 3 + 3 + 2 \times 1 \\ &= (3+3+3+3) + 3 + (1+1) \\ &= 3+3+3+3+3+1+1\end{aligned}$$

▷ **I.19.** *Euler's pentagonal number theorem.* To find a recurrence for the partition numbers, we consider the reciprocal of the generating function $P(z)$:

$$A(z) = \prod_{n=1}^{\infty}(1-z^n)$$

Euler's Pentagonal Number Theorem states that this product expands into the following sparse power series:

$$\begin{aligned}A(z) &= \sum_{k=-\infty}^{\infty}(-1)^k z^{k(3k+1)/2} \\ &= \sum_{k=-\infty}^{\infty}(-1)^k z^{k(3k-1)/2} \\ &= 1 - z^1 - z^2 + z^5 + z^7 - z^{12} - z^{15} + \cdots\end{aligned}$$

The exponents $g_k = \frac{k(3k-1)}{2}$ are the generalized pentagonal numbers. The product of $P(z)$ and $A(z)$ must equal 1:

$$\sum_{n=0}^{\infty} p(n)z^n \cdot \sum_{k=-\infty}^{\infty} (-1)^k z^{g_k} = 1$$

Comparing the coefficient of z^n on both sides yields the algorithmically useful recurrence:

$$p(n) = p(n-1) + p(n-2) - p(n-5) - p(n-7) + p(n-12) + p(n-15) \cdots$$

A Python implementation is provided below:

```
def compute_partitions(N):
P = [0] * (N + 1)
P[0] = 1  # Base case: p(0) = 1

for i in range(1, N + 1):
    k = 1
    while True:
        # Calculate pentagonal numbers for k and -k
        g_k_pos = k * (3 * k - 1) // 2
        g_k_neg = k * (3 * k + 1) // 2

        # Determine sign: +1 for k=1,2 (terms 1,2), -1 for k=3,4 (terms 3,4)...
        # Simplest way: (-1)^(k-1)
        sign = 1 if k % 2 == 1 else -1

        # Add p(i - g_k_pos)
        if g_k_pos <= i:
            P[i] += sign * P[i - g_k_pos]
        else:
            break # No more terms will fit

        # Add p(i - g_k_neg)
        if g_k_neg <= i:
            P[i] += sign * P[i - g_k_neg]
        else:
            break

        k += 1
return P
```

▷ **I.20.** *A digital surprise.*

$$\begin{aligned}
\varphi &= (1-10^{-1})(1-10^{-2})(1-10^{-3})(1-10^{-4})\cdots \\
&= \left[\prod_{n=1}^{\infty}(1-z^n)\right]_{z=.1} \\
&= \left[1-z^1-z^2+z^5+z^7-z^{12}-z^{15}+z^{22}+z^{26}-\cdots\right]_{z=.1} \\
&= .\dot{9}-.1-.01+.00001+.0000001-.000000000001-\cdots
\end{aligned}$$

The partial sums are:

- $\varphi_1 = .9999999999999999999999999$
- $\varphi_2 = .8999999999999999999999999$
- $\varphi_3 = .8899999999999999999999999$
- $\varphi_4 = .8900099999999999999999999$
- $\varphi_5 = .8900100999999999999999999$

- $\varphi_6 = .890010099998999999999999$
- $\varphi_7 = .890010099998998999999999$
- $\varphi_8 = .890010099998999000000999$

▷ **I.21.** *Lattice points.* We are interested in the number of integer solutions to the inequality $\sum_{i=1}^{d} x_i^2 \leq n^2$. First, consider a single coordinate $x_i \in \mathbb{Z}$. The "size" contributed by this coordinate to the sum of squares is x_i^2. The generating function for the possible values of a single coordinate, where the exponent of z represents the square of the coordinate, is:

$$\Theta(z) = \sum_{x \in \mathbb{Z}} z^{x^2} = 1 + 2\sum_{n=1}^{\infty} z^{n^2}$$

Since each coordinate $x_1, \cdot, x_d$ is chosen independently, the generating function for the sum of squares $\sum x_i^2$ is the d-th power of the individual generating function:

$$S(z) = (\Theta(z))^d = \sum_{k=0}^{\infty} r_d(k) z^k$$

In this series, the coefficient $[z^k]S(z)$ (denoted as $r_d(k)$) is exactly the number of ways to represent the integer k as a sum of d squares. Geometrically, this is the number of lattice points on the sphere (the boundary) of radius $\sqrt{k}$. The number of points $N(n, d)$ in the closed ball of radius n is:

$$N(n, d) = \sum_{k=0}^{n^2} r_d(k)$$

In generating functions, the prefix sum of a sequence's coefficients is obtained by multiplying the generating function by $\frac{1}{1-z}$. Therefore:

$$\sum_{k=0}^{\infty} \left(\sum_{j=0}^{k} r_d(j) \right) z^k = \frac{1}{1-z} S(z) = \frac{1}{1-z} (\Theta(z))^d$$

To find the total count up to $k = n^2$, we simply extract the coefficient of z^{n^2}:

$$N(n, d) = [z^{n^2}] \frac{1}{1-z} (\Theta(z))^d$$

1.4 Words and regular languages

▷ **I.23.** *Alice, Bob, and coding bounds.* Following Proposition I.4, the class $\mathcal{C}$ of binary strings avoiding the pattern 11 has the OGF:

$$C(z) = \frac{1+z}{1-z-z^2}$$

To account for a variable-length coding scheme where codewords are selected from $\bigcup_{j=0}^{L} \mathcal{C}_j$, we consider the sum of the first $L+1$ coefficients. This is represented by the OGF:

$$S(z) = \frac{C(z)}{1-z} = \frac{1+z}{(1-z)(1-z-z^2)}$$

The asymptotic growth of the sum $\sum_{j=0}^{L} C_j$ is governed by the smallest singularity of $S(z)$. Since the pole at $z = 1/\varphi$ (where $\varphi = \frac{1+\sqrt{5}}{2}$) is closer to the origin than the pole at $z = 1$, the exponential growth is dominated by $C(z)$. Specifically, as $L \to \infty$:

$$\sum_{j=0}^{L} C_j \sim A \cdot \varphi^L$$

for some constant A. To find the required code length $L(n)$ for n bits of information, we solve for $2^n \leq \sum_{j=0}^{L} C_j$ by taking the base-2 logarithm:

$$n \leq \log_2 \left(\sum_{j=0}^{L} C_j \right) = L \log_2 \varphi + O(1)$$

Rearranging for L, we obtain the lower bound:

$$L \geq \frac{n}{\log_2 \varphi} + O(1) \approx 1.440420n + O(1)$$

$\triangleright$ **I.27.** *Congruence languages.* The DFA $\mathcal{A}$ for binary strings congruent to 2 (mod 7) is structured as follows:

- **States**: A set $\mathcal{Q} = \{S_0, \ldots, S_6\}$, where each state S_r represents the current remainder r modulo 7.
- **Initial State**: S_0 (representing the value 0).
- **Accepting State**: $\overline{Q} = \{S_2\}$.
- **Alphabet**: $\mathcal{A} = \{0, 1\}$.

Each transition follows the rule $q_r \circ b = q_{(2r+b) \pmod 7}$:

Table 1.1: State Transition

Current State (Remainder)	Input 0 ($2r \pmod 7$)	Input 1 ($2r+1 \pmod 7$)
S_0	S_0	S_1
S_1	S_2	S_3
S_2	S_4	S_5
S_3	S_6	S_0
S_4	S_1	S_2
S_5	S_3	S_4
S_6	S_5	S_6

$\triangleright$ **I.29.** *The forward equations.* By a "last-letter decomposition", the set $\mathcal{M}_k$ of words leading to state q_k from the initial state q_0 satisfies the combinatorial relation:

$$\mathcal{M}_k \cong [\![k = 0]\!] \cdot \{\epsilon\} + \sum_{j,\alpha: q_j \circ \alpha = q_k} (\mathcal{M}_j \times \{\alpha\})$$

where $[\![\cdot]\!]$ is the Iverson bracket. This translates to a linear system of OGFs:

$$M_k(z) = \delta_{k,0} + z \sum_j M_j(z) T_{j,k}$$

In vector notation, with $\mathbf{M}(z)$ as a column vector and $\mathbf{e}_0$ the first basis vector, the system is a transposed version of the backward equations:

$$\mathbf{M}(z)^T = \mathbf{e}_0^T + z\mathbf{M}(z)^T T$$

$\triangleright$ **I.33.** *Waiting times in strings.* To illustrate the formulas, let's define a prefix language $\hat{\mathcal{L}}$ containing one word of length 2 and one word of length 3. Let $\hat{\mathcal{L}} = \{aa, bbb\}$.

1. Verification of Prefix Property: The word aa is not a prefix of bbb, and bbb is not a prefix of aa. Since no word is a strict prefix of another, this is a valid prefix language $\hat{\mathcal{L}}$.

2. The Generating Function $\hat{L}(z)$: The OGF is the sum of z raised to the power of each word's length: $\hat{L}(z) = z^2 + z^3$.

3. Probability: $\hat{L}(1/2)$: This represents the probability that a random infinite string starts with a word in $\mathcal{L}$: $\hat{L}(1/2) = (1/2)^2 + (1/2)^3 = 1/4 + 1/8 = 3/8$. This means there is a 37.5% chance that a random string begins with a word from this language.

4. Expected Time: $\frac{1}{2}\hat{L}'(1/2)$. To find the expected time (or more precisely, the expected "matched length") at which a word in $\mathcal{L}$ is first encountered: $\hat{L}'(z) = 2z + 3z^2 \implies \hat{L}'(1/2) = 1.75 \implies E = 0.875$.

5. Breakdown of the Result: The total expected value of 0.875 is formed as follows:

   - Word "aa": Probability 1/4 of length 2 $\to$ Contribution: $2 \times 0.25 = 0.5$.
   - Word "bbb": Probability 1/8 of length 3 $\to$ Contribution: $3 \times 0.125 = 0.375$.
   - No Match: Probability 5/8 of length 0 $\to$ Contribution: $0 \times 0.625 = 0$.

$\triangleright$ **I.34.** *A probabilistic paradox on strings.* Consider a binary pattern of length k. The generating function $T(z)$ for the first occurrence of this pattern is given by:

$$T(z) = \frac{z^k}{z^k + (1 - mz)c(z)} = \frac{z^k}{z^k + (1 - 2z)c(z)}$$

The average waiting time (or expected occurrence time) can be derived directly from $T(z)$ by evaluating its derivative at the point of the alphabet's inverse size ($z = 1/2$):

$$\left[z\frac{d}{dz}T(z)\right]_{z=1/2} = 2^k c(1/2)$$

1.5 Tree structures

▷ **I.39.** *Unary–binary trees and Motzkin numbers.* We identify the characteristic polynomial as $\phi(u) = 1 + u + u^2$. To find the number of trees with n nodes, $M_n = [z^n]M(z)$, we apply the Lagrange Inversion Theorem:

$$\begin{aligned}
M_n &= \frac{1}{n}[u^{n-1}]\phi(u)^n \\
&= \frac{1}{n}[u^{n-1}](1+u+u^2)^n \\
&= \frac{1}{n}[u^{n-1}]\sum_j \binom{n}{j}(u+u^2)^j \\
&= \frac{1}{n}\sum_j \binom{n}{j}[u^{n-1}]u^j(1+u)^j \\
&= \frac{1}{n}\sum_j \binom{n}{j}[u^{n-j-1}]\sum_l \binom{j}{l}u^l \\
&= \frac{1}{n}\sum_j \binom{n}{j}\binom{j}{n-j-1} \\
&= \frac{1}{n}\sum_k \binom{n}{n-k}\binom{n-k}{n-(n-k)-1} \\
&= \frac{1}{n}\sum_k \binom{n}{k}\binom{n-k}{k-1}
\end{aligned}$$

▷ **I.41.** *The Lagrangean form of Schröder's GF.* We want to prove the last equality

$$\frac{(-1)^{n-1}}{n}\sum_k (-2)^k \binom{n}{k+1}\binom{n+k-1}{k} = \frac{1}{n}\sum_{k=0}^{n-2}\binom{2n-k-2}{n-1}\binom{n-2}{k}$$

Multiplying by n, the statement is equivalent to

$$(-1)^{n-1}\sum_{k=0}^{n-1}(-2)^k\binom{n}{k+1}\binom{n+k-1}{k} = \sum_{k=0}^{n-2}\binom{2n-k-2}{n-1}\binom{n-2}{k}$$

Step 1. Re-index the left sum. Using $\binom{n}{k+1} = \binom{n}{n-1-k}$ and $\binom{n+k-1}{k} = \binom{n+k-1}{n-1}$, set $j = n-1-k$:

$$L = (-1)^{n-1}\sum_{j=0}^{n-1}(-2)^{n-1-j}\binom{n}{j}\binom{2n-2-j}{n-1} = 2^{n-1}\sum_{j=0}^{n-1}\left(-\frac{1}{2}\right)^j\binom{n}{j}\binom{2n-2-j}{n-1}$$

Step 2. Express as a coefficient extraction. Since $\binom{2n-2-j}{n-1} = [z^{n-1}](1+z)^{2n-2-j}$, we have

$$\begin{aligned}
L &= 2^{n-1}\sum_{j=0}^{n-1}\left(-\frac{1}{2}\right)^j\binom{n}{j}[z^{n-1}](1+z)^{2n-2-j} \\
&= 2^{n-1}[z^{n-1}]\sum_{j=0}^{n-1}\left(-\frac{1}{2}\right)^j\binom{n}{j}(1+z)^{2n-2-j} \\
&= 2^{n-1}[z^{n-1}](1+z)^{2n-2}\sum_{j=0}^{n-1}\binom{n}{j}\left(-\frac{1}{2(1+z)}\right)^j \\
&= 2^{n-1}[z^{n-1}](1+z)^{2n-2}\left(1-\frac{1}{2(1+z)}\right)^n \\
&= 2^{n-1}[z^{n-1}](1+z)^{2n-2}\frac{(1+2z)^n}{2^n(1+z)^n} \\
&= \frac{1}{2}[z^{n-1}](1+z)^{n-2}(1+2z)^n
\end{aligned}$$

Step 3. Reverse the polynomial. Let $P(z) = (1+z)^{n-2}(1+2z)^n$, of degree $2n-2$. The coefficient of z^{n-1} in $P(z)$ equals the coefficient of z^{n-1} in its reversed polynomial $z^{2n-2}P(1/z)$:

$$z^{2n-2}P(1/z) = z^{2n-2}\left(1+\frac{1}{z}\right)^{n-2}\left(1+\frac{2}{z}\right)^n = (z+1)^{n-2}(z+2)^n$$

Hence

$$L = \frac{1}{2}[z^{n-1}](1+z)^{n-2}(2+z)^n$$

Step 4. Express the right sum as a coefficient. Similarly, $\binom{2n-k-2}{n-1} = [z^{n-1}](1+z)^{2n-k-2}$, so

$$\begin{aligned}
R &= \sum_{k=0}^{n-2}\binom{2n-k-2}{n-1}\binom{n-2}{k} \\
&= \sum_{k=0}^{n-2}[z^{n-1}](1+z)^{2n-k-2}\binom{n-2}{k} \\
&= [z^{n-1}]\sum_{k=0}^{n-2}(1+z)^{2n-k-2}\binom{n-2}{k} \\
&= [z^{n-1}](1+z)^{2n-2}\sum_{k=0}^{n-2}\binom{n-2}{k}(1+z)^{-k} \\
&= [z^{n-1}](1+z)^{2n-2}\left(1+\frac{1}{1+z}\right)^{n-2} \\
&= [z^{n-1}](1+z)^{2n-2}\frac{(2+z)^{n-2}}{(1+z)^{n-2}} \\
&= [z^{n-1}](1+z)^n(2+z)^{n-2}
\end{aligned}$$

Step 5. Show equality of the two coefficient expressions. We need to show $\frac{1}{2}(1+z)^{n-2}(2+z)^n$ and $(1+z)^n(2+z)^{n-2}$ have the same coefficient of z^{n-1}. Their difference is

$$\begin{aligned}\Delta &= \frac{1}{2}[z^{n-1}](1+z)^{n-2}(2+z)^n - [z^{n-1}](1+z)^n(2+z)^{n-2} \\ &= \frac{1}{2}[z^{n-1}](1+z)^{n-2}(2+z)^{n-2}[(2+z)^2 - 2(1+z)^2] \\ &= \frac{1}{2}[z^{n-1}](1+z)^{n-2}(2+z)^{n-2}(2-z^2)\end{aligned}$$

We must prove $[z^{n-1}]f(z) = 0$, where

$$f(z) = (1+z)^{n-2}(2+z)^{n-2}(2-z^2)$$

Step 6. A symmetry argument. $f(z)$ is a polynomial of degree $2n-2$. Compute $f(2/z)$:

$$\begin{aligned}f(2/z) &= (1+2/z)^{n-2}(2+2/z)^{n-2}\left(2-(2/z)^2\right) \\ &= \left(\frac{2+z}{z}\right)^{n-2}\left(\frac{2(1+z)}{z}\right)^{n-2}\left(2-\frac{4}{z^2}\right) \\ &= \frac{(2+z)^{n-2}(1+z)^{n-2}2^{n-2}}{z^{2n-4}} \cdot \frac{-2(2-z^2)}{z^2} \\ &= -2^{n-1}\frac{f(z)}{z^{2n-2}}\end{aligned}$$

Hence $z^{2n-2}f(2/z) = -2^{n-1}f(z)$. Writing $f(z) = \sum_{k=0}^{2n-2} a_k z^k$, the coefficient of z^{n-1} corresponds to $k = n-1$. Equating coefficients gives $a_{n-1}2^{n-1} = -2^{n-1}a_{n-1} \implies a_{n-1} = 0$.

▷ **I.42.** *Faster determination of Schröder numbers.*

$$S(z) = \frac{1}{4}(1+z-\sqrt{1-6z+z^2})$$

Consider in general

$$p(z)S(z) + q(z) = \sqrt[k]{r(z)}$$

where $p(z)$, $q(z)$, and $r(z)$ are functions of z. Taking the derivative with respect to z,

$$p'(z)S(z) + p(z)S'(z) + q'(z) = \frac{1}{k}r(z)^{\frac{1}{k}-1}r'(z)$$

Multiplying both sides by $r(z)$,

$$p'(z)r(z)S(z) + p(z)r(z)S'(z) + q'(z)r(z) = \frac{1}{k}(p(z)S(z)+q(z))r'(z)$$

Rearranging,

$$p(z)r(z)S'(z) + (p'(z)r(z) - \frac{1}{k}p(z)r'(z))S(z) = \frac{1}{k}q(z)r'(z) - q'(z)r(z)$$

Plugging $p(z) = -4$, $q(z) = 1+z$, $k = 2$, and $r(z) = 1-6z+z^2$ into the equation and simplifying yields

$$(z^2-6z+1)S'(z) + (3-z)S(z) = 1-z$$

from which the recurrence follows.

▷ **I.43.** *Fast determination of the Cayley–Pólya numbers.*

$$H(z) = z\exp\left(H(z) + \frac{1}{2}H(z^2) + \cdots\right)$$

Step 1. Take the Natural Logarithm. First, we take the log of both sides to bring down the terms in the exponent:

$$\ln H(z) = \ln z + \sum_{k=1}^{\infty} \frac{1}{k} H(z^k)$$

Step 2. Differentiate with respect to z. Now, we apply the derivative operator $\frac{d}{dz}$:

$$\frac{H'(z)}{H(z)} = \frac{1}{z} + \sum_{k=1}^{\infty} \frac{1}{k} H'(z^k) \cdot kz^{k-1} = \frac{1}{z} + \sum_{k=1}^{\infty} z^{k-1} H'(z^k)$$

Step 3. Multiply by $zH(z)$. To clear the fractions and align the powers of z, we multiply the entire equation by $zH(z)$:

$$zH'(z) = H(z)\left(1 + \sum_{k=1}^{\infty} z^k H'(z^k)\right)$$

Step 4. Substitute Power Series. We represent $H(z)$ as a formal power series $\sum_{n=1}^{\infty} H_n z^n$. Note that:

- $zH'(z) = \sum_{n=1}^{\infty} nH_n z^n$
- $z^k H'(z^k) = \sum_{m=1}^{\infty} mH_m z^{mk}$

Plugging these into our equation:

$$\sum_{n=1}^{\infty} nH_n z^n = H(z) + H(z)\sum_{k=1}^{\infty}\sum_{m=1}^{\infty} mH_m z^{mk}$$

Step 5. The Recurrence Relation. By extracting the coefficient of z^n from both sides, we arrive at the recurrence. The double sum represents a convolution where we look for indices such that $mk + j = n$ (where j is the index from the $H(z)$ term). The standard form for the recurrence of these coefficients is:

$$(n-1)H_n = \sum_{l=1}^{n-1} H_{n-l}\left(\sum_{d|l} dH_d\right)$$

Or, more commonly expressed by isolating H_n:

$$H_n = \frac{1}{n-1}\sum_{l=1}^{n-1} H_{n-l}\left(\sum_{d|l} dH_d\right) = \frac{1}{n-1}\sum_{l=1}^{n-1} a_l H_{n-l}$$

To manually calculate the first few values using this recurrence, we start with the base case $H_1 = 1$ representing the root of the tree.

- To find H_2, we set $n = 2$. The summation runs for $l = 1$. The only divisor of 1 is 1. $a_1 = \sum_{d|1} dH_d = 1 \cdot H_1 = 1 \implies H_2 = \frac{1}{2-1}(H_{2-1} \cdot a_1) = 1$.
- For H_3, the summation runs for $l = 1$ and $l = 2$. The divisors of 2 are 1 and 2. $a_2 = \sum_{d|2} dH_d = 1 \cdot H_1 + 2 \cdot H_2 = 3 \implies H_3 = \frac{1}{3-1}(H_{3-1} \cdot a_1 + H_{3-2} \cdot a_2) = 2$.
- For H_4, the summation runs for $l = 1, 2, 3$. The divisors of 3 are 1 and 3. $a_3 = \sum_{d|3} dH_d = 1 \cdot H_1 + 3 \cdot H_3 = 7 \implies H_4 = \frac{1}{4-1}(H_{4-1} \cdot a_1 + H_{4-2} \cdot a_2 + H_{4-3} \cdot a_3) = 4$.

▷ **I.50.** *Excursions, bridges, and meanders.* A meander of final altitude k consists of k "up-steps" ($\mathcal{Z}$) that lead to new "floors", with Dyck paths (excursions $\mathcal{D}$) occurring between them and at the very end.

$$\mathcal{M} = \mathcal{D} + (\mathcal{D} \times \mathcal{Z} \times \mathcal{D}) + (\mathcal{D} \times \mathcal{Z} \times \mathcal{D} \times \mathcal{Z} \times \mathcal{D}) + \cdots = \mathcal{D} \times \sum_{k=0}^{\infty} (\mathcal{Z} \times \mathcal{D})^k$$

A bridge is a sequence of primitive excursions. A primitive excursion starts at 0, takes one step (up or down), wanders without returning to 0, and then takes one step back to 0.

- **Going Above:** Starts with +1, ends with -1, with a Dyck path in between: $\mathcal{Z} \times \mathcal{D} \times \mathcal{Z}$
- **Going Down:** Starts with -1, ends with 1, with mirrored a Dyck path in between: $\mathcal{Z} \times \mathcal{D} \times \mathcal{Z}$

$$\mathcal{B} = \text{SEQ}(\mathcal{Z} \times \mathcal{D} \times \mathcal{Z} + \mathcal{Z} \times \mathcal{D} \times \mathcal{Z}) = \text{SEQ}(2 \times \mathcal{Z}^2 \times \mathcal{D})$$

We want to show $M_n = \binom{n}{\lfloor n/2 \rfloor}$. A meander represents a path that stays in the first quadrant. Let $N(n, k)$ be the number of paths of length n ending at altitude k. By the reflection principle, the number of paths that hit -1 and end at $k \geq 0$ is exactly the same as the number of unrestricted paths that end at $-k-2 < -1$. Thus, Meanders end at $k =$ (Paths that end at k) $-$ (Paths that hit -1 and end at k) $=$ (Paths that end at k) $-$ (Paths end at $-k-2$). Summing over all possible final altitudes $k \geq 0$, we get a telescoping sum:

$$M_n = \sum_{k \geq 0} [N(n, k) - N(n, k + 2)]$$

- n is even ($n = 2m$): $M_n = N(n, 0) = N(2m, 0) = \binom{2m}{\frac{2m+0}{2}} = \binom{2m}{m} = \binom{n}{\lfloor n/2 \rfloor}$
- n is odd ($n = 2m+1$): $M_n = N(n, 1) = N(2m+1, 1) = \binom{2m+1}{\frac{2m+1+1}{2}} = \binom{2m+1}{m} = \binom{n}{\lfloor n/2 \rfloor}$

▷ **I.17.** *The complexity of boolean functions.*

$$D_n = O(16^n m^n n^{-3/2}) \implies \sum_{j=0}^{v} D_j = O(16^v m^v v^{-3/2})$$

We prove the following general statement.
Let $c > 0$ and $\alpha > 0$. Define $a_n = c^n n^{-\alpha}$ for $n \geq 1$ and let $a_0 > 0$ be a constant. If $D_n = O(a_n)$, then $\sum_{j=0}^{v} D_j = O(c^v v^{-\alpha})$ as $v \to \infty$.

Proof. Because $D_n = O(a_n)$, there exists $M > 0$ and $N \geq 1$ such that $D_n \leq Mc^n n^{-\alpha}$ for all $n \geq N$. For $v \geq N$ we split

$$\sum_{j=0}^{v} D_j \leq \underbrace{\sum_{j=0}^{N-1} D_j}_{C_0} + M \sum_{j=N}^{v} c^j j^{-\alpha}$$

Focus on $S(v) = \sum_{j=N}^{v} c^j j^{-\alpha}$. Split at $j = \lfloor v/2 \rfloor$:

$$S(v) = \underbrace{\sum_{j=N}^{\lfloor v/2 \rfloor} c^j j^{-\alpha}}_{\text{early}} + \underbrace{\sum_{j=\lfloor v/2 \rfloor+1}^{v} c^j j^{-\alpha}}_{\text{tail}}$$

Early terms. For $j \leq v/2$ we have $c^j \leq c^{v/2}$. Also since $\alpha > 0$, the factor $j^{-\alpha}$ is decreasing; its maximum is at $j = N$, so $j^{-\alpha} \leq N^{-\alpha}$. Thus

$$\sum_{j=N}^{\lfloor v/2 \rfloor} c^j j^{-\alpha} \leq N^{-\alpha} \sum_{j=N}^{\lfloor v/2 \rfloor} c^j \leq N^{-\alpha} \frac{c^{v/2+1}}{c-1} = O(c^{v/2})$$

Because $c > 1$, $c^{v/2}$ is exponentially smaller than c^v, so this part is $o(c^v v^{-\alpha})$.
Tail terms. For $j \geq v/2$, $j^{-\alpha} \leq (v/2)^{-\alpha} = 2^\alpha v^{-\alpha}$. Consequently

$$\sum_{j=\lfloor v/2 \rfloor+1}^{v} c^j j^{-\alpha} \leq 2^\alpha v^{-\alpha} \sum_{j=\lfloor v/2 \rfloor+1}^{v} c^j \leq 2^\alpha v^{-\alpha} \frac{c^{v+1}}{c-1} = O(c^v v^{-\alpha})$$

Putting it together, $S(v) = o(c^v v^{-\alpha}) + O(c^v v^{-\alpha}) = O(c^v v^{-\alpha})$. Hence

$$\sum_{j=0}^{v} D_j = C_0 + MS(v) = O(1) + O(c^v v^{-\alpha}) = O(c^v v^{-\alpha})$$

Now we show

$$\sum_{j=0}^{v} D_j = O(16^v m^v v^{-3/2}) = o(\|\Omega\|)$$

by taking for v the value

$$v = \frac{2^m}{4 + \log_2 m}$$

Note that

$$v \log_2 16m = \frac{2^m}{4 + \log_2 m}(4 + \log_2 m) = 2^m \implies 16^v m^v = 2^{2^m}$$

Hence the whole bound becomes

$$\sum_{j=0}^{v} D_j = O(2^{2^m} v^{-3/2})$$

Dividing by $\|\Omega\|$,

$$\frac{\sum_{j=0}^{v} D_j}{\|\Omega\|} = O(v^{-3/2})$$

As $m \to \infty$, $v \to \infty$, so $v^{-3/2} \to 0$; therefore

$$\frac{\sum_{j=0}^{v} D_j}{\|\Omega\|} \to 0$$

which is exactly $\sum_{j=0}^{v} D_j = o(\|\Omega\|)$.

1.6 Additional constructions

▷ **I.62.** *Sequences without repeated components.* A sequence without repeated components is essentially an ordered arrangement of a subset of distinct elements from a base class $\mathcal{B}$. To derive its generating function, we first consider the construction of a set of distinct elements (the power set), then account for the ordering. The generating function for the power set construction $\mathrm{PS{\small ET}}(\mathcal{B})$ is given by:

$$A(z,u) = \prod_{\beta\in\mathcal{B}} \left(1 + uz^{|\beta|}\right) = \exp\left(\sum_{j\geq 1}(-1)^{j-1}\frac{u^j}{j}B(z^j)\right)$$

Here, the coefficient $[u^k]A(z,u) = f^{\langle k\rangle}(z)$ represents the generating function for sets containing exactly k distinct components. While the power set construction satisfies the "no-repeat" constraint, it treats the collection as unordered. To transform these sets into sequences, we multiply the generating function for sets of size k by $k!$ to account for all possible permutations. Using the Eulerian integral representation $k! = \int_0^\infty e^{-u}u^k du$, we can express the generating function for distinct sequences, $\mathcal{S}_{\text{distinct}}(z)$, as follows:

$$\begin{aligned}
\mathcal{S}_{\text{distinct}}(z) &= \sum_{k\geq 0}(\text{Sets of size k})\cdot k! \\
&= \sum_{k\geq 0} f^{\langle k\rangle}(z)\cdot k! \\
&= \sum_{k\geq 0} f^{\langle k\rangle}(z)\cdot \int_0^\infty e^{-u}u^k du \\
&= \sum_{k\geq 0} \int_0^\infty f^{\langle k\rangle}(z)e^{-u}u^k du \\
&= \int_0^\infty \sum_{k\geq 0} f^{\langle k\rangle}(z)e^{-u}u^k du \\
&= \int_0^\infty \sum_{k\geq 0} f^{\langle k\rangle}(z)u^k \cdot e^{-u} du \\
&= \int_0^\infty A(z,u)\cdot e^{-u} du \\
&= \int_0^\infty \exp\left(\sum_{j\geq 1}(-1)^{j-1}\frac{u^j}{j}B(z^j)\right) e^{-u} du
\end{aligned}$$

1.7 Perspective

Chapter 2

LABELLED STRUCTURES AND EXPONENTIAL GENERATING FUNCTIONS

2.1 Labelled classes

2.2 Admissible labelled constructions

2.3 Surjections, set partitions, and words

2.4 Alignments, permutations, and related structures

2.5 Labelled trees, mappings, and graphs

▷ **II.19.** *Labelled hierarchies.*

$$L = z + e^L - 1 - L \implies L(z) = T\left(\frac{1}{2}e^{z/2-1/2}\right) + \frac{z}{2} - \frac{1}{2}$$

Step 1. Re-arranging the target equation. We start with the given equation, moving all terms involving L to the left side and constant/independent terms to the right:

$$2L + 1 - e^L = z$$

This form does not immediately match $T(z) = ze^{T(z)}$. To make them compatible, we should attempt to isolate an exponential term that mirrors the functional form of the tree function.

Step 2. Utilizing the Lambert W identity. The Cayley tree function $T(z)$ is related to the **Lambert W function** (specifically $W(-z)$). The identity $T(z) = ze^{T(z)}$ can be rewritten as:

$$-T(z)e^{-T(z)} = -z$$

Which implies $T(z) = -W(-z)$.

Step 3. Substitution and Solving. Let's rearrange the original equation to isolate the exponential:

$$e^L = 2L - z + 1$$

For equations where a variable appears both linearly and in the exponent, the solution is typically logarithmic or involves the W function. Let's rewrite the equation for L in the form $f(L)e^{f(L)} =$ something:

$$1 = (2L - z + 1)e^{-L}$$

Now, we want the term inside the parentheses to match the exponent. Let u be the linear part related to L. If we aim for $u = \frac{z-1}{2} - L$, then:

$$2u = z - 1 - 2L$$

Substituting this back into our equation:

$$1 = (-2u)e^{-L}$$

Since $u = \frac{z-1}{2} - L$, we have $-L = u - \frac{z-1}{2}$. Substituting this into the exponent:

$$1 = -2u \cdot e^{u - \frac{z-1}{2}} = -2u \cdot e^u \cdot e^{\frac{1-z}{2}}$$

Now, isolate ue^u:

$$ue^u = -\frac{1}{2}e^{\frac{z-1}{2}}$$

Since $ue^u = v$ implies $u = W(v)$, and we know $W(v) = -T(-v)$:

$$u = W\left(-\frac{1}{2}e^{\frac{z-1}{2}}\right) = -T\left(\frac{1}{2}e^{\frac{z-1}{2}}\right)$$

Substituting u back in:

$$\frac{z-1}{2} - L = -T\left(\frac{1}{2}e^{\frac{z-1}{2}}\right)$$

Solving for L:

$$L = T\left(\frac{1}{2}e^{\frac{z-1}{2}}\right) + \frac{z-1}{2}$$

II.5.2. ***Mappings and associated graphs***. Let $\mathcal{T}$ be the class of all Cayley trees. Then

$$F = (1-T)^{-1} \implies F_n = n^n$$

This result follows from Lagrange inversion. The standard formulation states that

$$z = T/\phi(T) \implies [z^n]T(z) = \frac{1}{n}[w^{n-1}]\phi(w)^n$$

More generally,

$$[z^n]H(T(z)) = \frac{1}{n}[w^{n-1}]H'(w)\phi(w)^n$$

Substituting $H(w) = (1-w)^{-1}$ and $\phi(w) = e^w$ yields

$$\begin{aligned}
F_n &= n![z^n]F(z) \\
&= n!\frac{1}{n}[w^{n-1}]\left(\frac{1}{1-w}\right)'(e^w)^n \\
&= (n-1)![w^{n-1}]\frac{e^{nw}}{(1-w)^2} \\
&= (n-1)![w^{n-1}]\sum_{k=0}^{\infty}(k+1)w^k \cdot \sum_{j=0}^{\infty}\frac{n^j}{j!}w^j \\
&= (n-1)!\sum_{j=0}^{n-1}((n-1-j)+1)\frac{n^j}{j!} \\
&= (n-1)!\sum_{j=0}^{n-1}(n-j)\frac{n^j}{j!} \\
&= (n-1)!\left(\sum_{j=0}^{n-1} n\frac{n^j}{j!} - \sum_{j=0}^{n-1} j\frac{n^j}{j!}\right) \\
&= (n-1)!\left(\sum_{j=0}^{n-1}\frac{n^{j+1}}{j!} - \sum_{j=1}^{n-1}\frac{n^j}{(j-1)!}\right) \\
&= (n-1)!\left(\sum_{j=0}^{n-1}\frac{n^{j+1}}{j!} - \sum_{j=0}^{n-2}\frac{n^{j+1}}{j!}\right) \\
&= (n-1)!\frac{n^n}{(n-1)!} \\
&= n^n
\end{aligned}$$

Now consider $K = \log(1-T)^{-1}$. Then

$$K_n = n^{n-1}Q(n)$$

where

$$Q_n = 1 + \frac{n-1}{n} + \frac{(n-1)(n-2)}{n^2} + \cdots$$

From Note II.18 we have

$$n![z^n]\frac{T(z)^k}{k!} = \binom{n-1}{k-1}n^{n-k} \implies [z^n]\frac{T(z)^k}{k} = (k-1)!\binom{n-1}{k-1}n^{n-k} = \frac{(n-1)!}{(n-k)!}n^{n-k}$$

Consequently,

$$\begin{aligned} K_n &= n![z^n]K(z) \\ &= n![z^n]\sum_{j=1}^{\infty}\frac{T(z)^j}{j} \\ &= \sum_{j=1}^{\infty} n![z^n]\frac{T(z)^j}{j} \\ &= \sum_{j=1}^{\infty}\frac{(n-1)!}{(n-j)!}n^{n-j} \\ &= n^{n-1}\sum_{j=1}^{\infty}\frac{(n-1)!}{(n-j)!n^{j-1}} \\ &= n^{n-1}\left(1+\frac{n-1}{n}+\frac{(n-1)(n-2)}{n^2}+\cdots\right) \end{aligned}$$

II.5.2. ***Constrained mappings***. The class of all maps without fixed points has EGF

$$\frac{e^{-T(z)}}{1-T(z)}$$

We aim to show that

$$n![z^n]\frac{e^{-T(z)}}{1-T(z)} = (n-1)^n$$

Substituting $H(w) = \frac{e^{-w}}{1-w}$ and $\phi(w) = e^w$ into Lagrange inversion yields

$$\begin{aligned} n![z^n]\frac{e^{-T(z)}}{1-T(z)} &= n!\frac{1}{n}[w^{n-1}]\left(\frac{e^{-w}}{1-w}\right)'(e^w)^n \\ &= (n-1)![w^{n-1}]\frac{we^{(n-1)w}}{(1-w)^2} \\ &= (n-1)![w^{n-2}]\frac{e^{(n-1)w}}{(1-w)^2} \\ &= (n-1)![w^{n-2}]\sum_{k=0}^{\infty}(k+1)w^k\cdot\sum_{j=0}^{\infty}\frac{(n-1)^j}{j!}w^j \\ &= (n-1)!\sum_{j=0}^{n-2}((n-2-j)+1)\frac{(n-1)^j}{j!} \\ &= (n-1)!\sum_{j=0}^{m-1}(m-j)\frac{m^j}{j!} \\ &= (n-1)!\frac{m^m}{(m-1)!} \\ &= (n-1)!\frac{(n-1)^{n-1}}{(n-2)!} \\ &= (n-1)^n \end{aligned}$$

II.5.3. ***Unrooted trees and acyclic graphs***. To understand why $U(z) = \int_0^z T(w)\frac{dw}{w}$, we begin with the exponential generating functions:

$$T(z) = \sum_{n\geq 0} T_n \frac{z^n}{n!}, \quad U(z) = \sum_{n\geq 0} U_n \frac{z^n}{n!}$$

Differentiate $U(z)$ and then multiply by z (noting that $T_0 = 0$) gives:

$$z\frac{d}{dz}U(z) = z\frac{d}{dz}\sum_{n\geq 0} U_n \frac{z^n}{n!} = z\sum_{n\geq 1} nU_n \frac{z^{n-1}}{n!} = \sum_{n\geq 1} T_n \frac{z^n}{n!} = T(z)$$

Rearranging yields $\frac{d}{dz}U(z) = \frac{T(z)}{z}$. Integrating both sides with respect to z and using w as a dummy variable for the integration from 0 to z, we obtain:

$$U(z) - U(0) = \int_0^z \frac{T(w)}{w} dw$$

The EGF for rooted trees, $T(z)$, satisfies the functional equation:

$$T(z) = ze^{T(z)}$$

which can be rearranged to $z = Te^{-T}$. Differentiating with respect to T gives

$$dz = \left(e^{-T} - Te^{-T}\right) dT = e^{-T}(1-T)dT$$

Substituting $z = Te^{-T}$ and dz into the integral $U(z) = \int \frac{T(z)}{z} dz$ yields

$$U(z) = \int \frac{T}{Te^{-T}} \cdot e^{-T}(1-T)dT = \int (1-T)dT = T - \frac{T^2}{2} + C$$

Since $U(0) = 0$ and $T(0) = 0$, the constant $C = 0$.

$\triangleright$ **II.22.** *2–Regular graphs.* We establish a bijection between clouds and 2-regular graphs on n labeled vertices. Number the lines $1, 2, ..., n$.

- Line i in the cloud $\leftrightarrow$ vertex i in the graph.
- Intersection point $\{i,\ j\}$ in the cloud $\leftrightarrow$ edge $i - j$ in the graph.

Let n_i be the number of chosen intersection points on line i. Since each point lies on two lines, $\sum_{i=1}^n n_i = 2n$. *If no three points are collinear*, then $n_i \leq 2$ for all i, so $\sum_{i=1}^n n_i \leq 2n$. Equality $\sum_{i=1}^n n_i = 2n$ forces $n_i = 2$ for every i, i.e. every vertex has degree 2 in the graph.

Conversely, if every vertex has degree 2, then $n_i = 2$ for all i, so each line contains exactly two intersection points — hence no three collinear.

Thus the cloud condition and the 2-regular graph condition are equivalent.

$\triangleright$ **II.23.** *Wright's surgery.*

The Surgery in Forward Direction: Deconstruction

We start with a connected graph G of excess $k+1$. Pick an edge of G and an orientation (a direction) for it. This gives us a starting vertex A and an ending vertex B. Remove this directed edge from the graph. Depending on the role that edge played in the connectivity of G, one of two mutually exclusive cases occurs:

- **Case 1 (Connected)**: The removed edge was part of a cycle. The graph remains connected. Because we lost an edge but kept all vertices, the excess drops to k. The graph now possesses two designated, ordered, pointed vertices: a "source" point at A and a "target" point at B.
- **Case 2 (Disconnected)**: The removed edge was a bridge. The graph splits into two separate connected components, G_1 and G_2. If G_1 takes on an excess of h, then G_2 must take on the remaining excess of $k - h$. Furthermore, G_1 contains exactly one pointed vertex (where A used to connect), and G_2 contains exactly one pointed vertex (where B used to connect).

The Surgery in Backward Direction: Reconstruction
We want to reverse this process to build all possible directed-edge-pointed graphs of excess $k + 1$. We do this by reversing both cases:

- **Reversing Case 1**: Take a single connected graph of excess k. Pick any two distinct vertices in an ordered fashion, call them A and B, and draw a directed edge from A to B. If A and B already had an edge between them, adding a new one creates an illegal multi-edge. We must forbid picking pairs of vertices that are already directly connected.
- **Reversing Case 2**: Take two completely independent connected components — one of excess h and one of excess $k - h$. Pick one vertex in the first component (A) and one vertex in the second component (B). Draw a directed edge from A to B, merging them into a single connected graph of excess $k + 1$.

Now let's align these actions with the differential recurrence:

$$2\partial_y w_{k+1} = \left(z^2\partial_z^2 w_k - 2y\partial_y w_k\right) + \sum_{h=-1}^{k+1} (z\partial_z w_h) \cdot (z\partial_z w_{k-h})$$

Left-Hand Side: The Initial Pool

- w_{k+1}: Represents the universe of connected graphs of excess $k + 1$.
- ∂_y: Differentiates with respect to the edge-marker y. Combinatorially, $\frac{\partial}{\partial y}(y^E) = E \cdot y^{E-1}$, which means "choose one of the E edges and remove it."
- 2: Accounts for the orientation of the removed edge. Since an unoriented edge has 2 endpoints, choosing a direction means deciding which endpoint is the source (A) and which is the target (B).
- Together ($2\partial_y w_{k+1}$): Represents a graph of excess $k+1$ with one structurally deleted, but memory-tracked, directed edge.

Right-Hand Side, Term 1: The Connected Outcome (Case 1)

- w_k: The base graph of excess k.
- ∂_z^2: Differentiates with respect to the vertex-marker z twice. This selects an ordered pair of distinct vertices (A and B).

- z^2: Multiplies z^2 back in, ensuring that the total vertex count of the graph remains unchanged after the selection.
- $-2y\partial_y w_k$: The exclusion rule. $y\partial_y w_k$ selects an existing edge in the excess-k graph. That existing edge has 2 possible orientations (endpoints ordered as (A, B) or (B, A)). We subtract this out to prevent the $z^2\partial_z^2$ operator from choosing an existing edge's endpoints and accidentally creating a double-edge.

Right-Hand Side, Term 2: The Disconnected Outcome (Case 2)

- w_h and w_{k-h}: The two separate, independent components. Their generating functions are multiplied together because the components are combinatorially disjoint.
- $z\partial_z$: The pointing operator for vertices ($z\frac{\partial}{\partial z}(z^V) = V \cdot z^V$).
  - Applied to the first component ($z\partial_z w_h$), it selects exactly one vertex (A) to act as the source bridgehead.
  - Applied to the second component ($z\partial_z w_{k-h}$), it selects exactly one vertex (B) to act as the target bridgehead.
- $\sum_{h=-1}^{k+1}$: Sums over all valid ways the total remaining excess k can be distributed between the two components. Because the simplest possible connected graph is a tree (which has an excess of $E - V = -1$), the lowest possible excess a component can hold is -1.

2.6 Additional constructions

▷ **II.28.** *Do two permutations generate the symmetric group*? We want to show

$$n!^2\pi_n = (n-1)!I_n$$

The EGF of pairs of permutations $\sum_{n\geq 0} n!^2\frac{z^n}{n!} = \sum_{n\geq 0} n!z^n$ happens to be the OGF $P(z)$ of all permutations. By the exponential formula, the EGF $C(z)$ of connected graphs satisfies:

$$C(z) = \log P(z) = \log\left(\sum_{n\geq 0} n!z^n\right)$$

Since every permutation can be uniquely decomposed into a sequence of indecomposable components, the OGF $I(z)$ of indecomposable permutations satisfies:

$$I(z) = 1 - \frac{1}{P(z)}$$

Let's find a differential relationship. Take the derivative of $C(z)$ with respect to z:

$$C'(z) = \frac{P'(z)}{P(z)} = P'(z)(1 - I(z)) = P'(z) - P'(z)I(z)$$

Now, let's extract the coefficient of z^{n-1} from both sides of this derivative equation.

Left-Hand Side: Since $C(z) = \sum_{n\geq 1}(n!^2\pi_n)\frac{z^n}{n!}$, its derivative is:

$$C'(z) = \sum_{n\geq 1}(n!^2\pi_n)\frac{z^{n-1}}{(n-1)!}$$

Thus, the coefficient of z^{n-1} is:

$$[z^{n-1}]C'(z) = \frac{n!^2\pi_n}{(n-1)!} = n \cdot n!\pi_n$$

Right-Hand Side: Differentiate both sides of $P(z)(1 - I(z)) = 1$ with respect to z:

$$P'(z)(1 - I(z)) - P(z)I'(z) = 0 \implies P'(z) - P'(z)I(z) = P(z)I'(z)$$

Now we just need to extract the coefficient of z^{n-1} from $P(z)I'(z)$:

$$[z^{n-1}](P'(z) - P'(z)I(z)) = [z^{n-1}]P(z)I'(z) = \sum_{k=1}^{n} kI_k(n-k)!$$

Let's denote this sum by S_n. We will show $S_n = I_{n+1}$. Extracting the coefficient of z^N from $P(z) = 1 + I(z)P(z)$ for $N \geq 1$ yields the recurrence (noting that $I_0 = 0$)

$$\sum_{k=1}^{N} I_k(N-k)! = N! \tag{2.1}$$

Use 2.1 with $N = n + 1$:

$$\sum_{k=1}^{n+1} I_k(n+1-k)! = (n+1)! \implies \sum_{k=1}^{n} I_k(n+1-k)! = (n+1)! - I_{n+1} \tag{2.2}$$

Expand the left hand side of 2.2

$$\begin{aligned}\sum_{k=1}^{n} I_k(n+1-k)! &= \sum_{k=1}^{n} I_k(n+1-k)(n-k)! \\ &= (n+1)\sum_{k=1}^{n} I_k(n-k)! - \sum_{k=1}^{n} kI_k(n-k)!\end{aligned}$$

By 2.1 with $N = n$:

$$\sum_{k=1}^{n} I_k(n-k)! = n!$$

Hence the left hand side of 2.2 equals

$$\sum_{k=1}^{n} I_k(n+1-k)! = (n+1)n! - S_n = (n+1)! - S_n$$

Substituting into 2.2 gives $S_n = I_{n+1}$.

Finally $n \cdot n!\pi_n = S_n \implies n!^2\pi_n = (n-1)!I_n$.

2.7 Perspective

Chapter 3

COMBINATORIAL PARAMETERS AND MULTIVARIATE GENERATING FUNCTIONS

3.1 An introduction to bivariate generating functions (BGFs)

3.2 Bivariate generating functions and probability distributions

3.3 Inherited parameters and ordinary MGFs

▷ **III.7.** *Largest summands in compositions.* Let M be the largest summand in a random composition. We want to show that for any $\epsilon > 0$, as $n \to \infty$,

$$\mathbb{P}\left((1-\epsilon)\log_2 n \leq M \leq (1+\epsilon)\log_2 n\right) \to 1$$

Upper bound: first moment method

Let $r_+ = \lceil (1+\epsilon)\log_2 n \rceil$. If $M \geq r_+$, then there exists at least one summand of size $\geq r_+$. The number of such summands is $S = \sum_{k=r_+}^{\infty} X_k$ where X_k is the number of summands of size exactly k. By Markov's inequality,

$$\mathbb{P}(M \geq r_+) = \mathbb{P}(S \geq 1) \leq \frac{\mathbb{E}[S]}{1} = \sum_{k=r_+}^{\infty} \mathbb{E}[X_k]$$

Proposition III.4 gives $E[X_k] = \frac{n-k+3}{2^{k+1}} \leq \frac{n}{2^k}$. Then (noting that $X_k = 0$ for $k > n$)

$$\mathbb{E}[S] \leq \sum_{k=r_+}^{n} \frac{n}{2^k} < \frac{n}{2^{r_+-1}} \leq \frac{n}{2^{(1+\epsilon)\log_2 n - 1}} = 2n^{-\epsilon} \to 0$$

Hence $\mathbb{P}(M > (1+\epsilon)\log_2 n) \to 0 \implies \mathbb{P}(M \leq (1+\epsilon)\log_2 n) \to 1$.

Lower bound: second moment method

We have a random composition of n, given by $n-1$ independent gaps, each a "cut" with probability 1/2. Fix $\epsilon > 0$ and set $r_- = \lfloor (1-\epsilon)\log_2 n \rfloor$. We want to prove $\mathbb{P}(M > r_-) \to 1$. To get a strict inequality we simply look for a summand of size at least $r = r_- + 1$. If we can show that with high probability there is a run of r consecutive 1's, then $M \geq r > r_-$.

Cut the n ones into $m = \lfloor \frac{n}{r} \rfloor$ disjoint blocks of length r, ignoring leftover 1's. Inside a block of length r there are $r-1$ internal gaps. The block contains **no cut** internally with probability $p = \left(\frac{1}{2}\right)^{r-1}$, independently for each block (the gaps of different blocks are disjoint and independent).

Define the indicator random variables

$$I_i = \begin{cases} 1 & \text{if block i has no internal cuts} \\ 0 & \text{otherwise} \end{cases}$$

for $i = 1, \cdots, m$. Let $Y = \sum_{i=1}^m I_i$ which is the number of blocks that are completely uncut internally. If $Y \geq 1$ then at least one block is a single summand of size at least r; hence $M \geq r > r_-$. Therefore $\mathbb{P}(M > r_-) \geq \mathbb{P}(Y > 0)$.

By Paley-Zygmund inequality (we'll give a short proof later)

$$\mathbb{P}(Y > 0) \geq \frac{(\mathbb{E}[Y])^2}{\mathbb{E}[Y^2]}$$

Now $Y \sim \text{Binomial}(m, p)$. The first moment $\mathbb{E}[Y] = mp = \lfloor \frac{n}{r} \rfloor \left(\frac{1}{2}\right)^{r-1}$. Since $r = \lfloor (1-\epsilon)\log_2 n \rfloor + 1 \leq (1-\epsilon)\log_2 n + 1$, we have

$$\left(\frac{1}{2}\right)^{r-1} \geq \left(\frac{1}{2}\right)^{(1-\epsilon)\log_2 n} = n^{-(1-\epsilon)}$$

Since $r \sim (1-\epsilon)\log_2 n$,

$$\mathbb{E}[Y] = mp \geq \lfloor \frac{n}{r} \rfloor n^{-(1-\epsilon)} \sim \frac{n^\epsilon}{(1-\epsilon)\log_2 n} \to \infty$$

The second moment $\mathbb{E}[Y^2] = Var(Y) + (\mathbb{E}[Y])^2$, $Var(Y) = mp(1-p) \leq mp = \mathbb{E}[Y]$. Plugging into Paley-Zygmund:

$$\mathbb{P}(Y > 0) \geq \frac{(\mathbb{E}[Y])^2}{\mathbb{E}[Y^2]} = \frac{(\mathbb{E}[Y])^2}{Var(Y) + (\mathbb{E}[Y])^2} \geq \frac{(\mathbb{E}[Y])^2}{\mathbb{E}[Y] + (\mathbb{E}[Y])^2} = \frac{1}{1 + \frac{1}{\mathbb{E}[Y]}} \to 1$$

Therefore $\mathbb{P}(M > r_-) \to 1 \implies \mathbb{P}(M \geq (1-\epsilon)\log_2 n) \to 1$.

Direct Proof of

$$\mathbb{P}(Y > 0) \geq \frac{(\mathbb{E}[Y])^2}{\mathbb{E}[Y^2]}$$

Start with the expectation of Y and split it according to whether $Y > 0$:

$$\mathbb{E}[Y] = \mathbb{E}[Y \cdot \mathbf{1}_{Y>0}] + \mathbb{E}[Y \cdot \mathbf{1}_{Y=0}] = \mathbb{E}[Y \cdot \mathbf{1}_{Y>0}]$$

Apply the Cauchy-Schwarz inequality:

$$\mathbb{E}[Y] \leq \sqrt{\mathbb{E}[Y^2]}\sqrt{\mathbb{E}[\mathbf{1}_{Y>0}]} = \sqrt{\mathbb{E}[Y^2]}\sqrt{\mathbb{P}(Y > 0)}$$

Because all terms are nonnegative, we can square both sides to obtain

$$(\mathbb{E}[Y])^2 \leq \mathbb{E}[Y^2] \cdot \mathbb{P}(Y > 0) \implies \mathbb{P}(Y > 0) \geq \frac{(\mathbb{E}[Y])^2}{\mathbb{E}[Y^2]}$$

3.4 Inherited parameters and exponential MGFs

3.5 Recursive parameters

3.6 Complete generating functions and discrete models

3.7 Additional constructions

3.8 Extremal parameters

3.9 Perspective